

Cloud Service Models



1. Infrastructure as a Service (IaaS)

Example: Amazon EC2, Azure Virtual Machines, Google Compute Engine

1. Provides virtualized computing resources over the Internet.
2. Offers virtual machines (VMs) instead of physical servers.
3. Users can install and manage their own operating systems.
4. Provides scalable storage services.
5. Supports virtual networking and IP management.
6. Resources can be increased or decreased on demand.
7. Follows a pay-as-you-go pricing model.
8. Eliminates the need to purchase physical hardware.
9. Provides backup and disaster recovery options.
10. Suitable for hosting websites, databases, and enterprise applications.
11. **Example:** A company hosts its website on Amazon EC2 instead of buying physical servers.

2. Platform as a Service (PaaS)

Example: Google App Engine, Heroku, Azure App Service

1. Provides a complete platform for application development.
2. Developers can focus on coding without managing hardware.
3. Includes development tools and frameworks.
4. Supports application testing and deployment.
5. Provides database management services.
6. Automatically handles software updates.
7. Supports collaboration among development teams.
8. Reduces development time and effort.
9. Offers built-in security and monitoring features.
10. Suitable for developing web and mobile applications.

Example: A developer uses Google App Engine to build and deploy a web application without managing servers.

3. Software as a Service (SaaS)

Example: Google Workspace, Microsoft 365, Salesforce

1. Provides ready-to-use software applications over the Internet.
2. Users can access applications through a web browser.
3. No software installation is required.
4. Supports automatic updates and maintenance.

5. Accessible from any device with an Internet connection.
6. Uses subscription-based pricing.
7. Reduces software management responsibilities.
8. Supports multiple users simultaneously.
9. Provides data backup and security features.
10. Suitable for email, collaboration, and business applications.

Example: Students use Google Docs online without installing any software on their computers.

IaaS = Rent a Server

PaaS = Rent a Development Platform

SaaS = Use Ready-Made Software